Building three-dimensional figures on an isometric grid

Drawing a solid honestly on flat paper — and reading it back from its three views.

LESSON 179–181 | 3 × 40 MIN | MATEMATIKA 6, §35 | S7 8.4 3-D SHAPES

LESSON CLOCK

30' | 30' | 30' | 25' | 5

Drawing on the isometric grid	30 min
Counting the cubes	30 min
Plan and elevations	30 min
From views back to the solid	25 min
Homework	5 min

LEARNING OBJECTIVES

- Draw a solid made of unit cubes on an isometric grid.
- Count the cubes in a stack, including the hidden ones.
- Find the surface area of a stack of unit cubes.
- Draw the plan and the two elevations, and rebuild a solid from them.

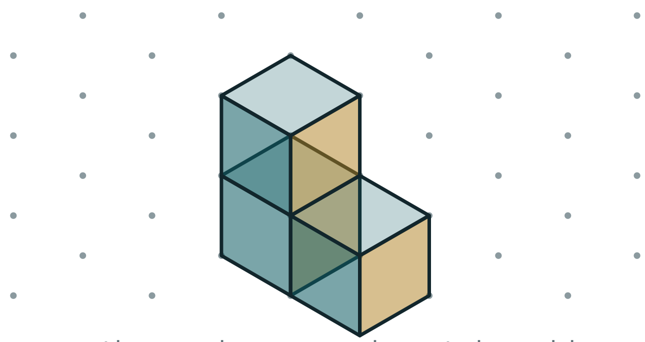
TEXTBOOK

National	pp. 512–521
Cambridge	Stage 7, pp. 88–94
Workbook	Exercise 8.5

01 Explanation

Drawing on the isometric grid

Isometric paper carries dots in triangles rather than squares. Vertical edges are drawn vertically; the other two directions run along the sloping lines of the grid. Every edge of a unit cube is then the same length on the page.



three cubes on an isometric grid

Three unit cubes drawn on an isometric grid

RULE	WHY
vertical edges stay vertical	height must look like height
the other two directions follow the grid lines	they keep equal lengths equal
never draw a horizontal line	there are none in this projection
draw the front cubes last	they must cover what is behind them

THE DRAWING IS NOT A PHOTOGRAPH

In a real photograph the far edges would look shorter. On an isometric grid they do not, which is exactly what makes the drawing measurable — and slightly strange to look at.

Counting the cubes

Count layer by layer, from the bottom up. A cube you cannot see still has to be there if something is resting on it.

STACK	CUBES	HIDDEN	SURFACE, IN UNIT SQUARES
an L of three cubes	*3*	*0*	*14*
$2 \times 2 \times 1$	*4*	*0*	*16*
$3 \times 2 \times 2$	*12*	*0*	*32*
$2 \times 2 \times 2$	*8*	*0*	*24*
$3 \times 3 \times 3$	*27*	*1*	*54*

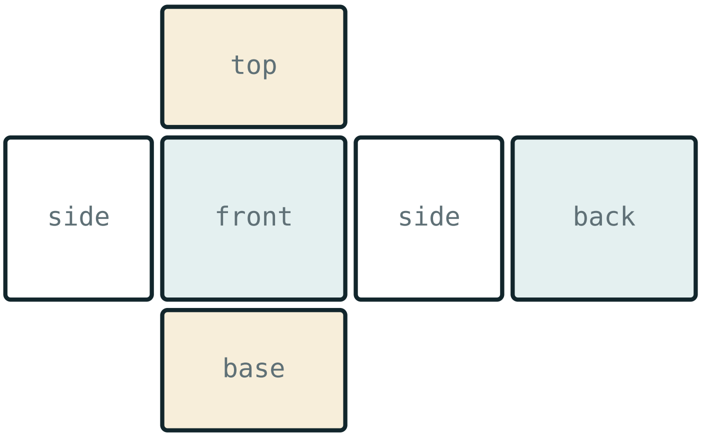
SURFACE AREA FROM THE JOINS

Three cubes have *18* faces in all. Two pairs are glued together, hiding *4* of them, so *14* are exposed. Every join hides exactly two faces — that is the quickest way to count.

Plan and elevations

Three flat drawings describe a solid without any perspective at all: the view from above and the views from the front and the side.

VIEW	LOOKING	SHOWS	FOR THE L OF THREE CUBES
plan	from above	length and depth	a *2* by *1* rectangle
front elevation	from the front	length and height	an L of three squares
side elevation	from the side	depth and height	a *1* by *2* rectangle



Flat views and flat nets both turn a solid into drawings on one page

EACH VIEW IS A SHADOW, NOT A PICTURE

Anything behind something else disappears. That is why the plan of the L is only two squares, even though the solid has three cubes.

From views back to the solid

Given three views, rebuild the solid layer by layer: the plan gives the footprint, the elevations give the height of each part.

PLAN	FRONT	SIDE	SOLID
2×2	2×2	2×2	a $2 \times 2 \times 2$ cube, 8 cubes
3×1	3×1	1×1	a row of 3 cubes
2×1	an L	1×2	the L of 3 cubes
3×1	a staircase	1×3	a staircase of 6 cubes

THREE VIEWS DO NOT ALWAYS FIX THE NUMBER OF CUBES

A cube tucked out of sight behind another changes none of the views. The views give the shape from outside; only the drawing or the solid itself gives the count for certain.

02 Worked examples

EXAMPLE 1 **A stack is 3 cubes long, 2 deep and 2 high. How many cubes, and what is its surface area in unit squares?**

1. $3 \cdot 2 \cdot 2 = 12$ cubes.
2. The three faces are 6, 4 and 6.
3. $2(6 + 4 + 6) = 32$ unit squares.
   It is a cuboid, so the old rule applies ✓

ANSWER

12 cubes, surface 32

EXAMPLE 2 **Three cubes form an L: two in a row, one on top of the left-hand cube. Find the exposed surface.**

1. $3 \cdot 6 = 18$ faces altogether.
2. Two joins, each hiding 2 faces: 4 hidden.
3. $18 - 4 = 14$ unit squares.
   Count the joins, not the faces ✓

ANSWER

14 unit squares

EXAMPLE 3 **Describe the plan and the two elevations of that L of three cubes.**

1. From above, the top cube hides the one below: a 2 by 1 rectangle.
2. From the front, heights 2 and 1: an L of three squares.
3. From the side, one deep and two high: a 1 by 2 rectangle.

   Each view is a shadow.

ANSWER

A 2×1 rectangle, an L, and a 1×2 rectangle

03 Interactive model

Build each solid from real cubes before drawing it; the hidden cube in a 3 × 3 × 3 stack is only convincing when the class has had to put it there.

Cubes, views and surfaces

A $2 \times 2 \times 2$ stack contains:

- 4
- 6
- 8
- 12

A $3 \times 3 \times 3$ stack has how many cubes no face of which can be seen?

- 0
- 1
- 3
- 9

Three cubes joined in an L expose:

- 18
- 16
- 14
- 12

A $2 \times 2 \times 1$ block has surface:

- 12
- 16
- 20
- 24

The plan is the view from:

- the front
- the side
- above
- below

The side elevation shows:

- length and depth
- depth and height
- length and height
- nothing

On isometric paper you never draw:

- a vertical line
- a horizontal line
- a sloping line
- a dot

Three views fix the number of cubes:

- always
- never
- not always
- only for cuboids

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Isometric grid	Izometrik to‘r	Изометрическая сетка
Unit cube	Birlik kub	Единичный куб
Stack	Ustma-ust taxlam	Стопка
Plan	Ustdan ko‘rinish	Вид сверху
Front elevation	Oldindan ko‘rinish	Вид спереди
Side elevation	Yon ko‘rinish	Вид сбоку
Hidden cube	Yashirin kub	Скрытый куб
Exposed face	Ochiq yoq	Открытая грань

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

On an isometric grid the dots form:

- squares
- triangles
- circles
- hexagons

A vertical edge of a cube is drawn:

- sloping left
- sloping right
- vertically
- horizontally

Each join between two cubes hides:

- 1 face
- 2 faces
- 4 faces
- nothing

A cube with something resting on it:

- need not be there
- must be there
- is optional
- is hollow

The front elevation shows:

- length and depth
- length and height
- depth and height
- volume

A view is:

- a photograph
- a shadow, with nothing behind
- a net
- a cross-section

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Cubes in a $2 \times 2 \times 1$ stack

   ANSWER 4

2. Cubes in a $2 \times 2 \times 2$ stack

   ANSWER 8

3. Cubes in a $3 \times 2 \times 2$ stack

   ANSWER 12

4. Cubes in a $3 \times 3 \times 3$ stack

   ANSWER 27

5. The dots of an isometric grid form

   ANSWER Triangles

6. A vertical edge is drawn

   ANSWER Vertically

7. The plan is the view from

   ANSWER Above

Medium · the standard question

7 PROBLEMS

1. The surface of a $2 \times 2 \times 1$ block

   ANSWER 16 unit squares

2. The surface of a $2 \times 2 \times 2$ block

   ANSWER 24 unit squares

3. Cubes in a staircase of 1, 2 and 3

   ANSWER 6

4. The exposed faces of an L of three cubes

   ANSWER 14

5. The front elevation of a $2 \times 2 \times 2$ stack

   ANSWER A 2 by 2 square

6. The plan of a $3 \times 2 \times 2$ stack

   ANSWER A 3 by 2 rectangle

7. Cubes with no visible face in a $3 \times 3 \times 3$ stack

   ANSWER 1

Hard · stretch and reason

7 PROBLEMS

1. The surface of a $3 \times 2 \times 2$ block

   ANSWER 32 unit squares

2. The surface of a $3 \times 3 \times 3$ block

   ANSWER 54 unit squares

3. The exposed faces of a staircase of $1, 2, 3$ cubes

   ANSWER 24

4. A plan, front and side all 2 by 2: the largest possible solid

   ANSWER A $2 \times 2 \times 2$ cube of 8 cubes

5. Why can three views not fix the number of cubes?

   ANSWER A hidden cube changes none of them

6. Which view shows depth and height?

   ANSWER The side elevation

7. How many faces does each join between two cubes hide?

   ANSWER 2

07 Homework

Homework — 5 tasks

Use isometric paper for the drawings and squared paper for the three views.

1. Draw a $2 \times 2 \times 2$ stack of cubes on isometric paper.
2. Draw the L of three cubes and write its plan, front and side elevations.
3. Find the number of cubes and the surface area of a $4 \times 2 \times 2$ stack.
4. Build a staircase of $1, 2, 3$ and 4 cubes and count the cubes.
5. Draw a solid of your own from six cubes and give its three views.